AUDIO STEM SEPERATION

Digital Signal Processing

TEAM 2

P Sajai Krishna(40
Niya Santhosh(39)
Sandra Pradeep(52)
Sunisha S(67)
Jeevan CS(16)

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INTRODUCTION



MECHANISM

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- Vocal Stem Isolation & Linear-Phase FIR Filter Design
- Treble Subband, Recombination Synthesis & Dual-Domain Validation

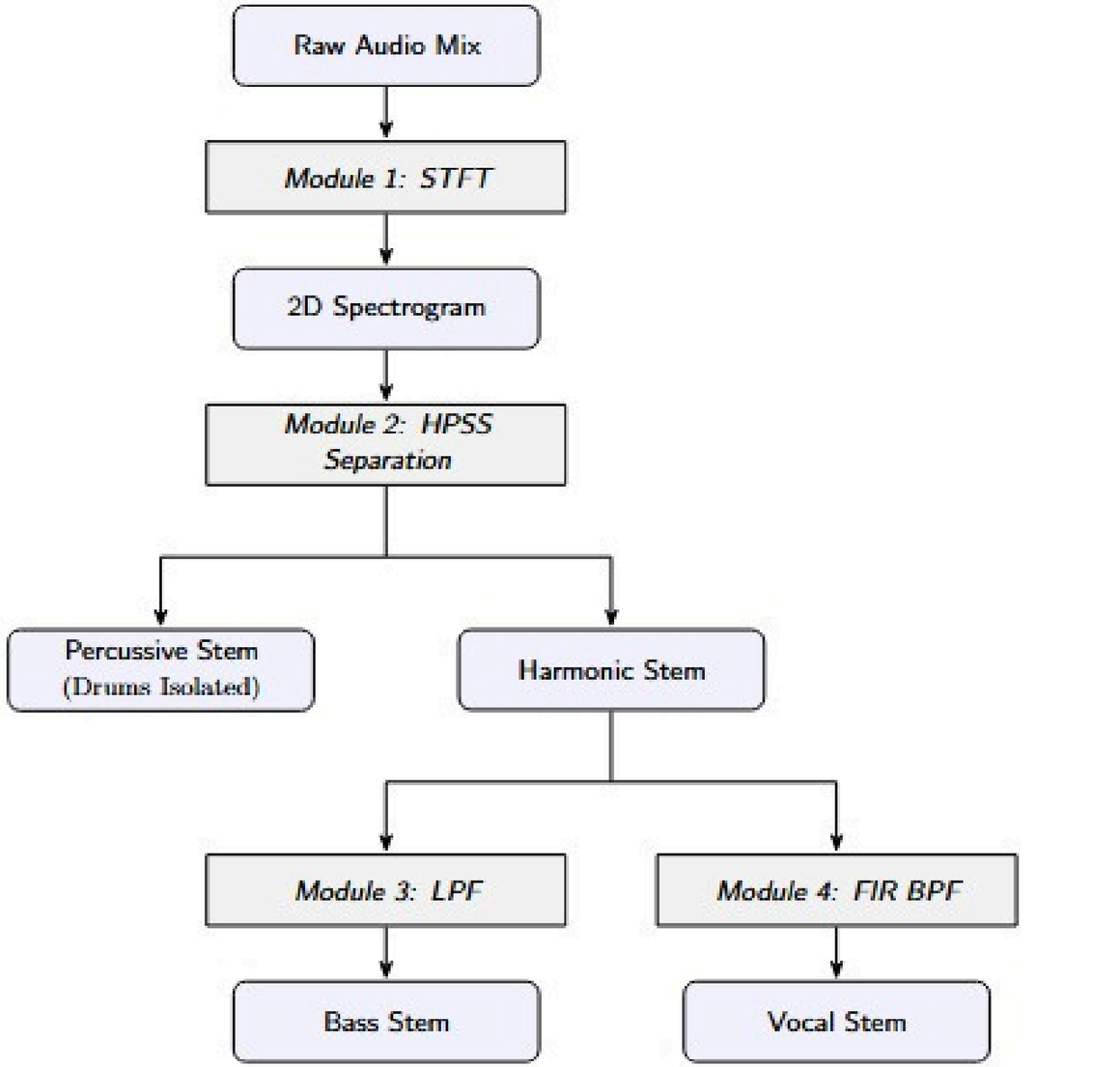


CONCLUSION

Intro

This project presents an end-to-end audio source separation pipeline that transforms a mixed audio signal into isolated bass, vocal, and treble components. The process begins with signal conditioning, normalization, and STFT time-frequency analysis to prepare the audio for spectral processing. It then applies 2D spectrogram anisotropic filtering and HPSS masking to separate harmonic and percussive content while preserving important time-frequency structures. Bass and vocal stems are extracted using IIR Butterworth and linear-phase FIR filter designs, ensuring smooth low-frequency separation and phase-consistent vocal isolation. Finally, the treble subband is isolated, all components are recombined, and the output is validated in both time and frequency domains to confirm separation quality and signal integrity.

Work Flow Execution Chart:



Signal Conditioning

Preparing the Input Audio
Load the mixed audio signal.u
Remove DC offset, if present.
Convert stereo to mono, if required.
Ensure a suitable sampling rate.

We perform signal conditioning. Our input is a mixed audio signal containing different sounds, such as bass, vocals, drums and other instruments.

Before analysing it, we prepare the audio. DC offset is an unwanted constant shift in the signal, so we remove it if present. We may also convert stereo audio into mono if a single channel is sufficient, and ensure that the sampling rate is suitable.

Signal Normalization

Standardizing Audio Amplitude
Scale the signal to a consistent amplitude range.
Prevent unnecessarily low or high signal levels.
Preserve relative sample amplitud
es.

After conditioning, we perform normalization. The input audio may have a low amplitude or a high amplitude, so normalization scales the signal such that its maximum absolute sample value becomes one

STFT Time-Frequency Analysis

Short-Time Fourier Transform
Divide audio into short, overlapping frames.
Apply a window function to each frame.
Compute the Fourier Transform of each frame.
Represent frequency content as it changes over time.

A normal waveform shows how the signal amplitude changes over time, but it does not clearly show which frequencies occur at each moment.
To solve this, STFT divides the audio into short, overlapping frames. A window function is applied to each frame, and the Fourier Transform is calculated. By repeating this process, we obtain frequency information for different time segments.
This information is represented using a spectrogram, where the horizontal axis represents time, the vertical axis represents frequency, and colour intensity represents the magnitude of the frequency components.

2D Spectrogram Anisotropic Filtering & HPSS Masking

Non-Stationary Time-Frequency Separation of Harmonic and Percussive Audio Stems

1

System Role:

- Primary non-linear separation stage in the DSP pipeline.

2

Input Specifications:

- Complex STFT tensor Z_xx(f, t)
- Real-valued magnitude spectrogram |Z_xx|

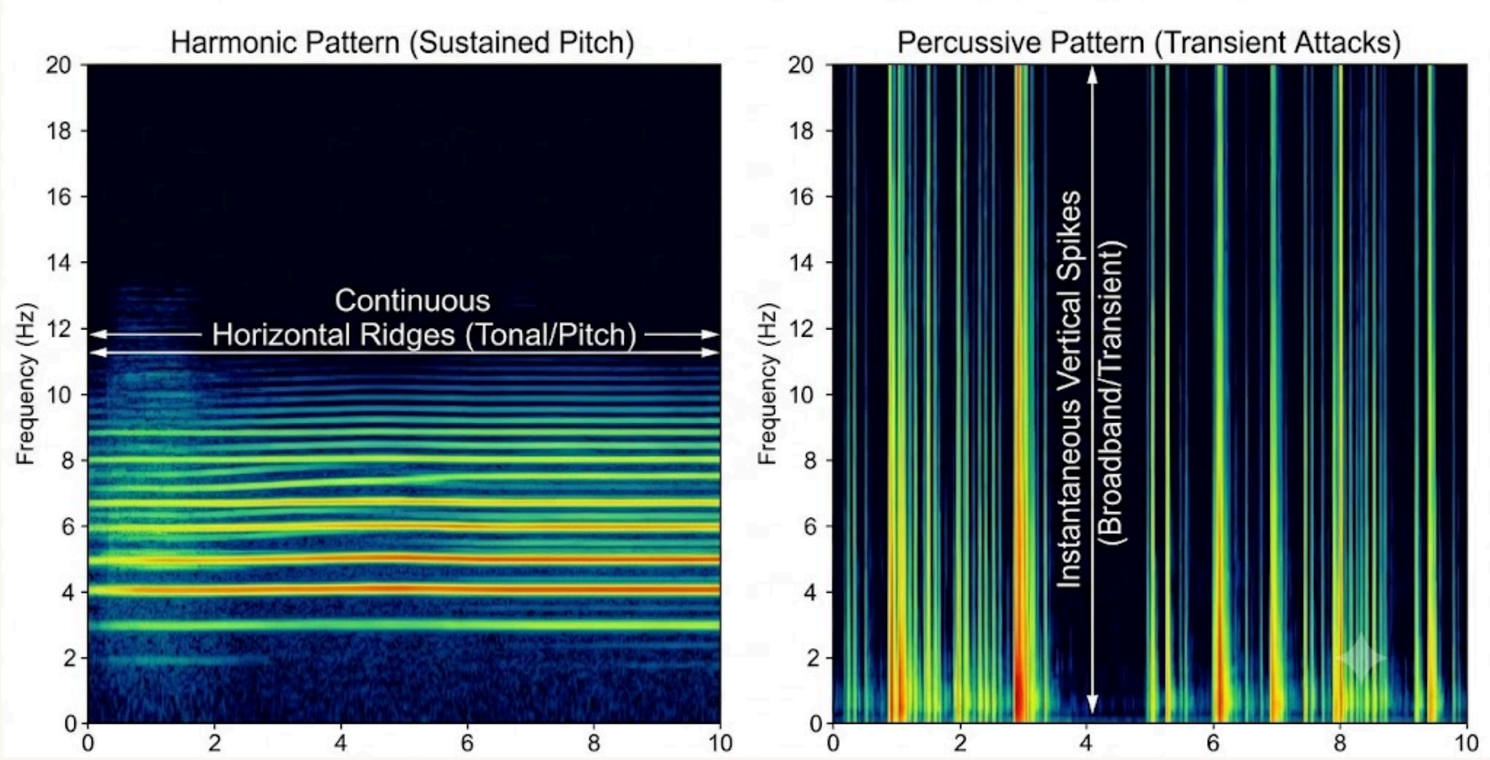
3

Output Deliverables:

- Percussive Stem Ydrum
- Harmonic Residual Yharmonic

2D Spectrogram Anisotropic Filtering & HPSS Masking

HPSS Time-Frequency Morphology (Horizontal vs. Vertical Signatures).



2D Spectrogram Anisotropic Filtering & HPSS Masking

Directional 2D Median Filter Kernels (Anisotropic Filtering)

- Horizontal Kernel (1, 31):

$$\mathbf{H}_{\mathbf{mag}}(f,t) = \mathbf{median}_{k \in [-15,15]} \left(|Z_{xx}(f, t+k)| \right)$$

- Vertical Kernel (31, 1):

$$\mathbf{P}_{\mathbf{mag}}(f,t) = \mathbf{median}_{m \in [-15,15]} \left(|Z_{xx}(f+m, t)| \right)$$

2D Spectrogram Anisotropic Filtering & HPSS Masking

HPSS masking:Soft Wiener-Style Masking & Energy Conservation

Mask Formulas:

$$M_H(f,t) = \frac{H_{\text{mag}}^2(f,t)}{H_{\text{mag}}^2(f,t) + P_{\text{mag}}^2(f,t) + \epsilon}$$

$$M_P(f,t) = \frac{P_{\text{mag}}^2(f,t)}{H_{\text{mag}}^2(f,t) + P_{\text{mag}}^2(f,t) + \epsilon}$$

- Partition of Unity:

$$M_H(f,t) + M_P(f,t) \approx 1$$

2D Spectrogram Anisotropic Filtering & HPSS Masking

Inverse STFT (ISTFT) Resynthesis

$$Z_P = M_P \odot Z_{xx}$$

$$y_{\text{drums}} = \text{iSTFT}(Z_P)$$

$$Z_H = M_H \odot Z_{xx},$$

$$y_{\text{harmonic}} = \text{iSTFT}(Z_H),$$

Bass Stem Isolation & IIR Butterworth Filter Theory

OBJECTIVE

Isolate bass and sub-bass content (below 250 Hz) from the harmonic stem using deterministic signal processing.

CORE METHODOLOGY

HPSS harmonic stem, then a 6th-order IIR Butterworth low-pass filter with zero-phase forward-backward filtering.

EVALUATION CRITERION

Dual-domain check: a 30 ms time-domain snapshot and log-magnitude frequency spectra.

Signal Ingestion & Profiling Metrics

Input Source

Harmonic stem y_harmonic from HPSS (2048-sample STFT window, 1024 overlap, soft mask M_H). The .wav is downmixed to mono via the channel mean.

Amplitude Normalization

Signal divided by its maximum absolute value, so the dynamic range is [-1.0, 1.0] and filtering stays free of clipping.

44.1 kHz

Sampling frequency (fs)

Bass Stem Extraction via IIR Butterworth Filter

1 The Acoustic Challenge

Bass sits below 250 Hz and must be split from the vocal band (300 Hz+) with no ripple.

2 The Solution

6th-order IIR Butterworth low-pass: steep roll-off from only 7 coefficients.

1

3 Cutoff Specification

fc = 250 Hz (-3 dB point), Wn = 250 / 22050 = 0.0113.

DSP Backend Implementation Details

Zero-Phase Processing (filtfilt)

- Forward and backward passes cancel phase distortion, $\theta(\omega) = 0°$
- Prevents comb-filtering when stems are recombined
- Magnitude is squared: effective order 12, -6 dB at cutoff

Complementary Subbands

- Bass: < 250 Hz, 6th-order Butterworth LPF
- Vocals: 300-3400 Hz, 401-tap FIR
- Treble: > 4500 Hz, 4th-order Butterworth HPF
- Drums: HPSS percussive stem

Dual-Domain Evaluation

Time domain

Smooth bass wave (about ±0.28) tracks the mix with no time skew.

Frequency domain

Peak near 200-250 Hz, then about 50dB down by 400 Hz.

Guard band

250-300 Hz gap separates bass from vocals.

Advanced Audio Demultiplexing: Vocal Stem Isolation & Linear-Phase FIR Design

A Rigorous Digital Signal Processing (DSP) Framework

Objective:
Isolate lead vocals and musical components from a complex polyphonic mixture using deterministic signal processing

Core Methodology:
Asymmetric hybrid filtering combined with 2D STFT spectrogram analysis.

Evaluation Criterion:
Dual-domain verification through time-domain waveforms and log-magnitude frequency spectra.

Advanced Audio Demultiplexing: Vocal Stem Isolation & Linear-Phase FIR Design



Signal Ingestion & Profiling Metrics

Input Source: Composite uncompressed .wav track downmixed to mono via (L+R)/2 to eliminate stereo panning discrepancies.

Amplitude Normalization: Dynamic range scaled to [-1.0, 1.0] to prevent numerical clipping during filtering stages.

Execution Profile (from script runtime)

Sampling Frequency (fs): 44100Hz

- Total Duration: 29.52seconds(1,301,807 samples)
- Peak Amplitude: 1.0000 | RMS Energy: 0.2726

Advanced Audio Demultiplexing: Vocal Stem Isolation & Linear-Phase FIR Design

Vocal Stem Extraction via Linear-Phase FIR Filter

The Acoustic Challenge

Human speech and vocal formants are extremely sensitive to phase dispersion and time smearing.

The Solution

401-tap Type-I Linear-Phase Finite Impulse Response (FIR) Band-Pass Filter.

Passband Specifications

Tuned strictly to 300 Hz - 3400 Hz (capturing primary speech formants F_1, F_2, F_3).

Windowing Function

Hamming window applied to control sidelobe leakage while maintaining strict coefficient symmetry ($h[n] = h[M-1-n]$).

Advanced Audio Demultiplexing: Vocal Stem Isolation & Linear-Phase FIR Design

DSP Backend Implementation Details

Filter Synthesis Code

firwin(num_taps=401, cutoff=[300.0, 3400.0], pass_zero='bandpass', fs=44100, window='hamming')

Zero-Phase Bidirectional Processing (filtfilt)

- Applies forward and backward passes to cancel phase distortion completely (theta(omega) = 0 degree)
- Prevents comb-filtering artifacts during final stem recombination.

Complementary Subbands

Paired with a 6th-order Butterworth LPF (<250 Hz) for bass and a 4th-order HPF (>4500 Hz) for treble.

SOLUTION OVERVIEW

Advanced Audio Demultiplexing: Vocal Stem Isolation & Linear-Phase FIR Design

Dual-Domain Evaluation

Time-Domain Snapshot (30 ms window at t = 2.0 s):
Displays phase-aligned vocal waves against the composite mix without temporal skew.

- **Frequency Spectrum (N_FFT = 8192, Hann Window, Delta_f_approx = 5.38 Hz):**
  - Confirms clean isolation inside the 300 Hz - 3400 Hz vocal formant corridor.
  - Demonstrates sharp stopband roll-off attenuating out-of-band signals by >40 dB.

Slide 6: Conclusion & Real

Treble Subband & Recombination Synthesis

1 Treble Stem Isolation

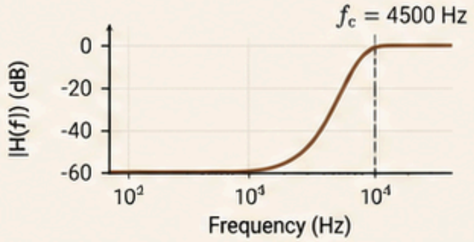
Extracting High-Frequency Components

- **Objective** : Extract treble components above 4500 Hz.
- **Filter** : 4th-order Butterworth High-Pass Filter (HPF).
- **Purpose** : Preserve cymbals, high-frequency harmonics and other treble details.
- **Input** : Harmonic component (from HPSS).
- **Output** : Treble stem $x_T[n]$.

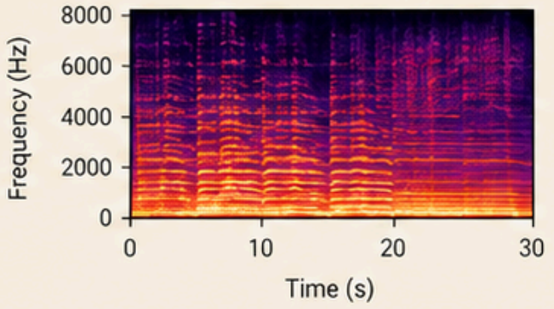
Butterworth High-Pass Filter

$$H(s) = \frac{s^4}{s^4 + \omega_c^4}$$

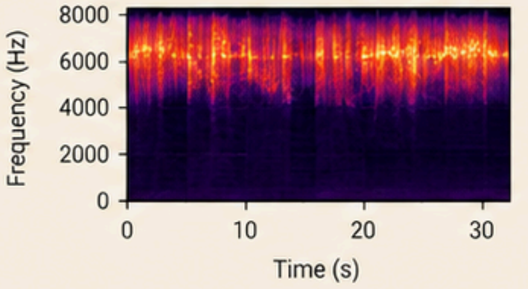
Order : 4
Cut-off frequency (f_c) : 4500 Hz
Type : High-Pass
Magnitude response:



Harmonic Spectrogram (Input)



Treble Stem Spectrogram (Output)



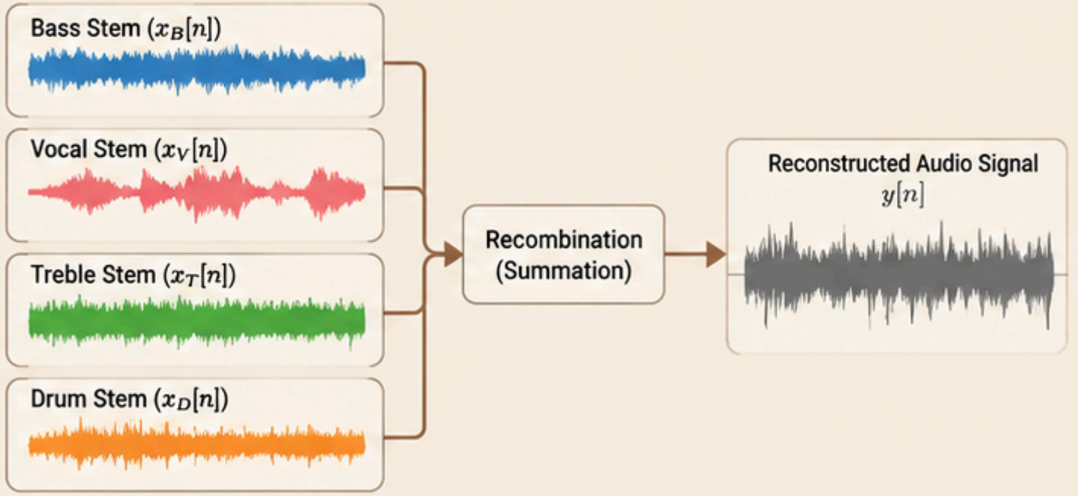
2 Recombination Synthesis

Reconstructing the Final Audio Signal

- Combine the separated stems to reconstruct the audio.
- Ensure proper time alignment and equal length of all stems.
- Maintains the overall structure and natural sound of the original signal.

$$y[n] = x_B[n] + x_V[n] + x_T[n] + x_D[n]$$

$x_B[n]$: Bass stem
$x_V[n]$: Vocal stem
$x_T[n]$: Treble stem
$x_D[n]$: Drum/percussive stem

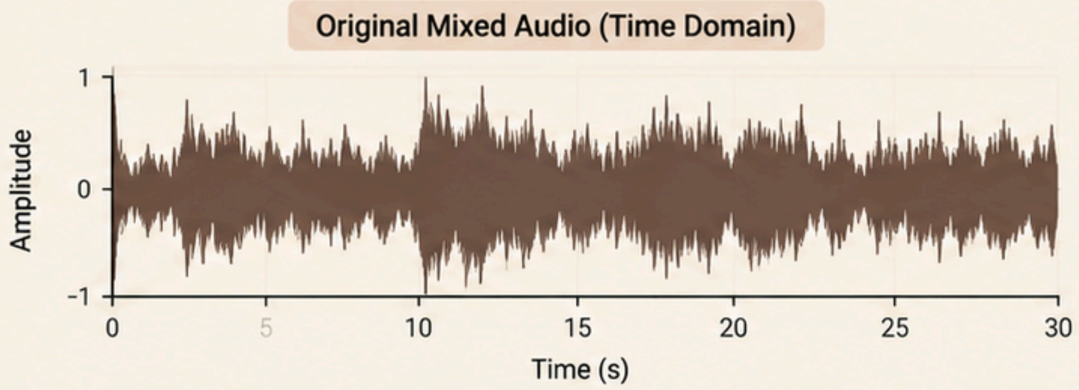


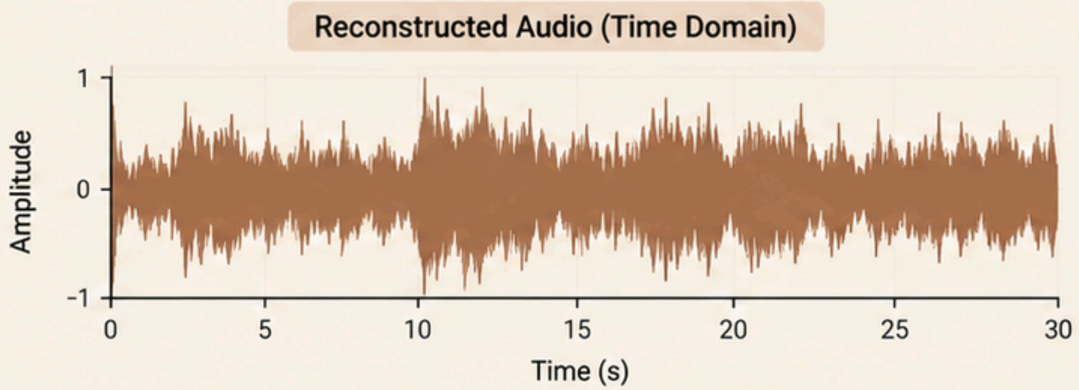
Note: Simple addition works best when the stems are designed to be complementary. Overlapping components can cause distortion in the reconstructed signal.

Dual-Domain Validation

1 Time-Domain Analysis

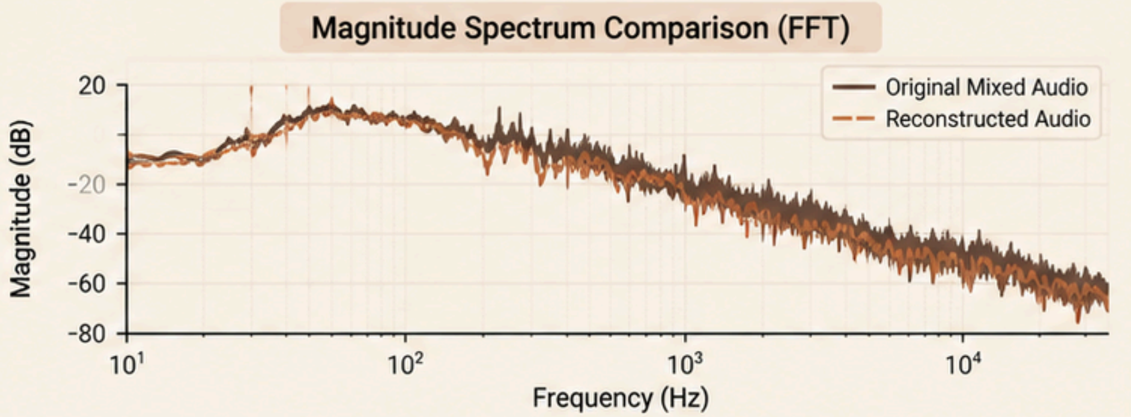
- Compare the original and reconstructed waveforms.
- Check amplitude consistency and timing alignment.
- Identify any clipping or reconstruction errors.





2 Frequency-Domain Analysis

- Compare the FFT magnitude spectra of the original and reconstructed signals.
- Verify that the frequency content is preserved.
- Identify any unwanted spectral leakage or distortion.



3 Performance Metrics

Mean Squared Error (MSE)

$$MSE = \frac{1}{N}\sum_{n=0}^{N-1}(x[n] - \hat{x}[n])^2$$

Signal-to-Noise Ratio (SNR)

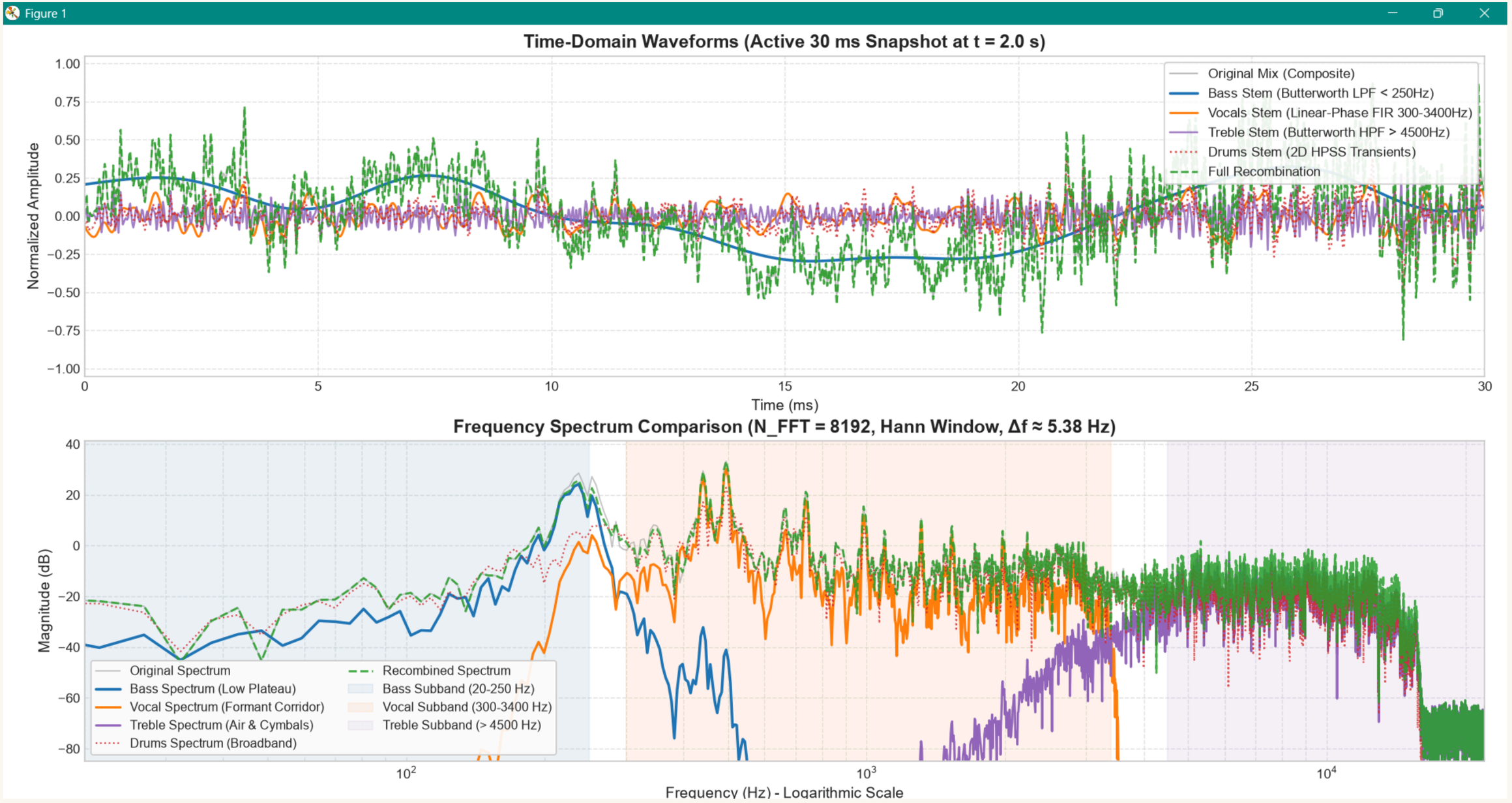
$$SNR = 10\log_{10}\left(\frac{\sum x[n]^2}{\sum (x[n] - \hat{x}[n])^2}\right)$$

Listening Test

- Perceptual audio quality
- Minimal artifacts
- Clear separation of stems

4 Final Outcome

Reconstructed audio with minimal distortion and preserved frequency characteristics.


Figure 1
Time-Domain Waveforms (Active 30 ms Snapshot at t = 2.0 s)
Normalized Amplitude
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20
25
30
Time (ms)
Original Mix (Composite)
Bass Stem (Butterworth LPF < 250Hz)
Vocals Stem (Linear-Phase FIR 300-3400Hz)
Treble Stem (Butterworth HPF > 4500Hz)
Drums Stem (2D HPSS Transients)
Full Recombination
Frequency Spectrum Comparison (N_FFT = 8192, Hann Window, Δf ≈ 5.38 Hz)
Magnitude (dB)
40
20
0
−20
−40
−60
−80
10^2
10^3
10^4
Frequency (Hz) - Logarithmic Scale
Original Spectrum
Bass Spectrum (Low Plateau)
Vocal Spectrum (Formant Corridor)
Treble Spectrum (Air & Cymbals)
Drums Spectrum (Broadband)
Recombined Spectrum
Bass Subband (20-250 Hz)
Vocal Subband (300-3400 Hz)
Treble Subband (> 4500 Hz)

Real-World Applications



1 Music Production

- Isolate vocals and instruments for remixing.
- Create karaoke tracks.
- Independent control during audio mastering and mixing.



2 Noise Reduction

- Separate speech from background music and ambient noise.
- Enhance voice clarity in calls and recordings.
- Useful for podcasts and online meetings.



3 Film & Video Editing

- Separate dialogue, background music and sound effects.
- Adjust each component independently.
- Improves flexibility in post-production.



4 Hearing Assistance

- Enhance speech by reducing competing background sounds.
- Useful for people with hearing difficulties.
- Can improve speech intelligibility in noisy environments.



5 AI & Audio Analysis

- Extract individual sound sources for speech recognition.
- Analyse music structure and instrument characteristics.
- Useful in machine learning datasets and content classification.

Practical Example: Karaoke System



- Remove the vocal stem from a mixed song to create an instrumental version.
- Preserves the background music (bass, drums, treble).
- Can be used for karaoke, remixing and music education.



Our Project's Contribution

Uses STFT, HPSS and digital filters to separate audio into bass, vocal, treble and percussive components for independent processing.

The quality of separation depends on the overlap between sound sources and the limitations of the filtering method.

Conclusion

- Successfully engineered a collaborative, multi-stage DSP system that isolatesmixed audio into clean Drums, Bass, Vocal, and Accompaniment stems.

- Practical Engineering Impact: Validated that classical, deterministic signal processing provides a lightweight, real-time audio separation framework without the high compute cost or latency of neural networks.

The End